Progress Report 1

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Reporting period: Sep 13, 2026 to Oct 12, 2026

Work Date/Hours Logs

Date	Number of Hours	Description of work done
Sep 13, 2026	*1*	Created the MedGuard GitHub repo and wrote the initial README
Sep 15, 2026	*2*	Set up Docker Compose (Zookeeper, Kafka, Kafka UI) and tested the container pipeline; added setup instructions for teammate
Sep 21, 2026	*1*	Wrote the IoMT device simulator and the initial FastAPI gateway /ingest endpoint
Sep 22, 2026	*0.5*	Updated README with run instructions
Sep 23, 2026	*2*	Wrote the project proposal document and checked it into DocumentsAndReports
Oct 1, 2026	*1*	Rewrote the README to cover the full stack and run steps
Oct 2, 2026	*2*	Added patient context to device records (patient registry, prescription checks); added training dataset loader and instructions (CICIDS2017, ECU-IoHT)
Oct 10, 2026	*2*	Merged teammate's TLS branch with patient-context, found the TLS flags dropped in the merge (issue #4), restored them, generated mkcert certs, wrote TLS setup docs, cleaned up .gitignore
Oct 11-12, 2026	*2*	Tested the full stack over HTTPS end to end and prepared the demo for Progress Report 1

Summary Description of work done during this reporting period

The work done so far has been setting up the bare bones of the project itself. The Docker and Kafka stack works for containerization and data sorting and directing. Producers and consumers are programs that use Kafka. Kafka stores and routes the messages so the data can be processed independently and scale under larger loads since this is hospital medical device data we're working with. The endpoints have been created for various reasons and the python files are grouped under endpoints. We have a dataset to simulate patent records as well as checks made in the gateway. It checks one shared bearer token, validates the schema, flags insecure configs, and flags prescription mismatches. `device_id` is self-reported, so a device can still claim to be another one. Patient registry has also been obtained from sample datasets. That's the intended workflow, for now I'm testing with hand written patient data. David wrote the TLS branch. I did the merges, and I found and fixed the lost flags (issue #4). I had problems with setting up docker as virtualization wasn't enabled in my bios. At least that's what I thought. Task Manager showed it enabled. What fixed it was the WSL update and the Windows hypervisor setting. And Kafka didn't work because of my version of python. The proposal didn't ask for patient data, but I added it for context.

Repo Check in of Implementation completed

The files and folders I have checked in the repo are as follows:

- docker-compose.yml: Zookeeper, Kafka, Kafka UI, gateway and database services
- gateway/: FastAPI gateway (main.py, schemas.py, auth.py, kafka_producer.py, vulnerability_rules.py, patient_registry.py, prescription_checks.py)
- ingestion/: device simulator (simulate_devices.py) and dataset instructions
- data/: patient registry (patients.json)
- ml/: dataset loader (load_dataset.py)
- DocumentsAndReports/: project proposal
- README.md: setup, run and TLS instructions

AI Use Section

Table of AI Tools and Specific Use

AI Tool Name	Version, Account Type	Specific feature for which the AI tool was used	Value Addition
Claude (Claude Code)	Sonnet 5.5, Free plan	Docker Compose and Kafka setup help; debugging Docker virtualization, build errors, and the kafka-python incompatibility on Python 3.12	*Had to figure out how Docker and Kafka work and how they can scale with more load. Read publications as well.*
Claude (Claude Code)	Sonnet 5.5, Free plan	Drafting the FastAPI gateway (bearer-token auth, Pydantic schemas, vulnerability and prescription-mismatch checks) and the device simulator	*Figured out what security rules are implemented concerning medical devices*
Claude (Claude Code)	Sonnet 5.5, Free plan	Drafting the patient registry feature and the CICIDS2017 / ECU-IoHT dataset loader	*Cleaning up datasets as well as feature selection*
Claude (Claude Code)	Sonnet 5.5, Free plan	Drafting the proposal and task-split documents; diagnosing the dropped TLS flags after the branch merge	*Fixing merge conflicts arising from TLS implementation.*

Appendix: Prompt History

```
[83] (2026-10-06) I'll need to send him these instructions so his Claude can work with that
so if you can make a small doc that can explain it for him and give enough context that his
Claude can do it please

[84] (2026-10-06) He's still working on stream preprocessing

[85] (2026-10-06) Q: What do you want for the stream preprocessing side?
A: Just help me understand what stream preprocessing actually needs to do

[86] (2026-10-08) Is the stream preprocessing entirely for Kafka or am I confused

[87] (2026-10-08) So it's just cleaning the dataset for the model then
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[88] (2026-10-08) Okay I get it. Cleaning and giving to Kafka

[89] (2026-10-08) Okay got it. He did the tls part in a new branch so I'll work on merging it with the latest branch tomorrow

[90] (2026-10-10) name of the branch is acc feature/tls-encryption

[91] (2026-10-10) @"/root/.claude/uploads/b5ac2320-3bf6-5876-b02d-fa87f03f4548/80339a65-Project_1__IoMT___Health_Data_Pipeline_Anomaly_Detection.pdf" @"/root/.claude/uploads/b5ac2320-3bf6-5876-b02d-fa87f03f4548/0794cfc5-attachment.txt" Continue with the task described in the conversation above. Your most recent mcp__claude-code-remote__list_repos call has not run yet; nothing is wrong with the tool or its input. Run it now from the beginning with the tools you have, without assuming any result, file or state from it, and use the working directory and file locations you have now rather than ones earlier steps assumed.

The files from earlier in this conversation are available at these paths:
/mnt/user-data/uploads/Project_1__IoMT___Health_Data_Pipeline_Anomaly_Detection.pdf
/mnt/user-data/uploads/IMG_1827.jpeg (image, shown earlier in this conversation)
/mnt/user-data/uploads/IMG_1834.jpeg (image, shown earlier in this conversation)
/mnt/user-data/uploads/attachment.txt
/mnt/user-data/uploads/1790051002519_image.png (image, shown earlier in this conversation)
Read them there (those copies are read-only — copy a file elsewhere to modify it). The images marked as shown earlier are not attached to this message again; every other file is also attached, at an @-mentioned path, to this message or the file-delivery messages just before it; if a listed path is missing, use that @-mentioned copy instead.

These files, uploaded to this conversation's code-execution sandbox earlier, have been copied here at the same paths they had before:
/mnt/user-data/uploads/4495_ProposalTemplate_1_.docx
They are not attached to this message: read them at those paths (the copies are read-only — copy a file elsewhere to modify it). If one of them is missing, say so plainly and ask the user to upload it again.

[92] (2026-10-10) I want to merge patient-context with that feature/tls

[93] (2026-10-10) i cant see his branchj on my vs code

[94] (2026-10-10) ive pushed but I'm trying to run and test everythng

[95] (2026-10-10) did the flags get dropped in the merge, it could be my fault if they were n the branch to begin with

[96] (2026-10-10) How do i use mkcert? its not a command on ym system

[97] (2026-10-10) wait I'm confused, how am i solvng the flags being gone. I've installed mkcerts what next

[98] (2026-10-10) the app is running

[99] (2026-10-10) python ingestion/simulate_devices.py --count 20
Posting to https://localhost:8000/ingest (anomaly chance: 5%). Ground truth logged to ground_truth_log.csv. Ctrl+C to stop.

[1] normal | CONN-ERROR | infusion-pump-002 | -
[2] normal | CONN-ERROR | cardiac-monitor-002 | -
[3] normal | CONN-ERROR | cardiac-monitor-001 | -
[4] normal | CONN-ERROR | infusion-pump-001 | -
[5] normal | CONN-ERROR | pulse-ox-001 | -

[100] (2026-10-10) docker compose logs gateway --tail 30
gateway | INFO: Will watch for changes in these directories: ['/app']
gateway | INFO: Uvicorn running on https://0.0.0.0:8000 (Press CTRL+C to quit)
gateway | INFO: Started reloader process [1] using WatchFiles
gateway | INFO: Started server process [8]
gateway | INFO: Waiting for application startup.
gateway | INFO: Application startup complete.

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gateway | INFO: 172.18.0.1:58628 - "GET /docs HTTP/1.1" 200 OK
gateway | INFO: 172.18.0.1:58628 - "GET /openapi.json HTTP/1.1" 200 OK

$env:REQUESTS_CA_BUNDLE = "$(mkcert -CAROOT)\rootCA.pem"
python ingestion/simulate_devices.py --count 20
it worked after running this command

[101] (2026-10-10) yeah make a short readme

[102] (2026-10-10) What's next for me then, cuz david is working on preprocessing rn and i
know his next task is the isolation forest model

[103] (2026-10-10) So dashboard that can be viewed is the next checkpoint

[104] (2026-10-10) for the progress report, a demo is necessary in a sense

[105] (2026-10-10) But what exactly would the small demo be showing? A risk score based on
thje model?
```